



CS502 - Fundamentals of Algorithms



CS502 - Fundamentals of Algorithms

(Mid-Term) 2021 GRAND Quiz 90 TO 100% Correct Solve (Alhamdulillah)

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https://www.youtube.com/channel/UCjboEX65C_yN7as5di1OuIQ/playlists Sadiq Ahmed

<https://www.youtube.com/channel/UC0AihThg8aPr5TcfqTUKWQ> Sonu Ilyas Mughal





CS502:Grand Quiz

Question # 1 of 30 (Start time: 11:31:00 AM, 27 June 2021)

Efficient algorithm requires less computational _____.

Select the correct option

- | | |
|----------------------------------|-------------------------|
| ☒ | memory and running time |
| ☐ | running time |
| ☐ | notation |
| ☐ | memory |





CS502:Grand Quiz

Question # 2 of 30 (Start time: 11:32:43 AM, 27 June 2021)

$8n^2 + 2n - 3$ will eventually exceed $c^2(n)$ no matter how large we make _____.

Select the correct option

☐	this equation
☐	$2n$
☐	n
☒	c^2





CS502:Grand Quiz

Question # 3 of 30 (Start time: 11:33:26 AM, 27 June 2021)

Using _____ algorithm, efficiency is not given much importance.

Select the correct option

- | | |
|----------------------------------|-------------|
| ☐ | Merge sort |
| ☐ | Greedy |
| ☒ | Brute Force |
| ☐ | Processing |





Question # 4 of 30 (Start time: 11:34:04 AM, 27 June 2021)

$f(n)$ and $g(n)$ are asymptotically equivalent. This means that they have essentially the same _____ for large n .

Select the correct option

☐	Results
☒	Growth rates
☐	Variables
☐	Size





Question # 5 of 30 (Start time: 11:34:35 AM, 27 June 2021)

In chain matrix multiplication, the order of the matrices_____.

Select the correct option

- | | |
|----------------------------------|--------------------|
| ☒ | can not be changed |
| ☐ | is reverse |
| ☐ | is equal |
| ☐ | can be changed |

CS502:Grand Quiz

Quiz Start Time: 11:31 AM, 27 June 2021

Question # 6 of 30 (Start time: 11:35:02 AM, 27 June 2021)

Total Marks

If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions

Select the correct option

- | | |
|----------------------------------|-----|
| ☐ | 2x3 |
| ☐ | 2x2 |
| ☒ | 3x3 |
| ☐ | 3x2 |

[Click to Save Answer & Move to Next Question](#)



CS502:Grand Quiz

Question # 7 of 30 (Start time: 11:35:33 AM, 27 June 2021)

Merge sort algorithm discussed in handouts contains _____.

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | 3 loops |
| ☐ | 2 loops |
| ☒ | 4 loops |
| ☐ | one loop only |

Click





Question # 8 of 30 (**Start time: 11:36:01 AM, 27 June 2021**)

In _____ we have to find rank of an element from given input.

Select the correct option

☐	Brute force technique
☐	Plane Sweep algorithm
☒	Selection problem
☐	Merge sort algorithm

Click to Submit





CS502:Grand Quiz

Question # 9 of 30 (Start time: 11:36:35 AM, 27 June 2021)

The worst case running time of quick sort algorithm _____.

Select the correct option

- | | |
|----------------------------------|-----------------------|
| ☐ | Cannot be quadratic |
| ☐ | Is always Exponential |
| ☒ | Is quadratic |
| ☐ | Is linear |

Click to Save





CS502:Grand Quiz

Question # 10 of 30 (Start time: 11:37:03 AM, 27 June 2021)

In quick sort algorithm, pivots form _____.

Select the correct option

- | | |
|----------------------------------|--------------------|
| ☒ | Binary Search Tree |
| ☐ | Graph |
| ☐ | Queue |
| ☐ | Stack |

Click





CS502:Grand Quiz

Question # 11 of 30 (Start time: 11:37:37 AM, 27 June 2021)

In the statement
"if $(P[i].x < P[j].x) \ \& \ (P[i].y < P[j].y)$ ",
the number of times elements of P are accessed is _____

Select the correct option

☒	4
☐	3
☐	1
☐	2





CS502:Grand Quiz

Question # 12 of 30 (Start time: 11:38:22 AM, 27 June 2021)

Al-Khwarizmi was a/an _____.

Select the correct option

☒	mathematician
☐	artist
☐	astronomer
☐	khalifah





CS502:Grand Quiz

Question # 13 of 30 (Start time: 11:39:16 AM, 27 June 2021)

Quick sort is based on _____ strategy.

Select the correct option

☐	Graph Theory
☐	Greedy approach
☒	Divide-and-Conquer
☐	Dynamic programming





CS502:Grand Quiz

Question # 14 of 30 (Start time: 11:39:53 AM, 27 June 2021)

In Sieve technique, we solve the problem _____

Select the correct option

- | | |
|----------------------------------|-----------------------------|
| ☐ | Using Merge Sort algorithm |
| ☐ | Non recursively |
| ☒ | In recursive manner |
| ☐ | Using Brute force technique |





CS502:Grand Quiz

Question # 15 of 30 (Start time: 11:40:25 AM, 27 June 2021)

Insertion sort is an efficient algorithm for sorting a _____ number of elements.

Select the correct option

☐	Extra large
☐	Mediam
☐	Large
☒	Small

Click to Save





CS502:Grand Quiz

Question # 16 of 30 (Start time: 11:41:15 AM, 27 June 2021)

After partitioning array in Quick sort, pivot is placed in a position such that

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | Pivot is the last element of array |
| ☐ | Values larger than pivot are on left and smaller than pivot are on right |
| ☒ | Values smaller than pivot are on left and larger than pivot are on right |
| ☐ | Pivot is the first element of array |

Click to Save





CS502:Grand Quiz

Question # 17 of 30 (Start time: 11:41:54 AM, 27 June 2021)

Approach of solving geometric problems by sweeping a line across the plane is called _____ sweep.

Select the correct option

☐	Line
☐	Cube
☒	Plane
☐	Box





Question # 18 of 30 (Start time: 11:42:24 AM, 27 June 2021)

Total Mark

You are given a knapsack that can carry a maximum weight of 90. There are 4 items with weights {50, 40, 60, 70} and values {90, 80, 100, 250}. What is the maximum value of the items you can carry using the knapsack?

Select the correct option

Reload Math Equation

- | | |
|----------------------------------|-----|
| ☐ | 190 |
| ☐ | 180 |
| ☐ | 350 |
| ☒ | 170 |

[Click to Save Answer & Move to Next Question](#)





Question # 19 of 30 (Start time: 11:42:52 AM, 27 June 2021)

In Heap Sort algorithm, the maximum levels an element can move upward is _____.

Select the correct option

☐	Big-oh(log n)
☒	Theta (log n)
☐	O (1) i.e. Constant time
☐	Omega (log n)

Click to Save A





CS502:Grand Quiz

Question # 20 of 30 (Start time: 11:43:21 AM, 27 June 2021)

An algorithm is a mathematical entity, which is independent of _____.

Select the correct option

☐	Compiler and Programming language
☒	Programming language, Compiler and Machine
☐	Machine and Programming language
☐	Programming language





Question # 21 of 30 (Start time: 11:43:44 AM, 27 June 2021)

If we have 6 metrics in chain matrix multiplication problem then the number of table entries must be?

Select the correct option

- | | |
|----------------------------------|----|
| ☐ | 12 |
| ☒ | 36 |
| ☐ | 25 |
| ☐ | 30 |

Click to Save Ans





CS502:Grand Quiz

Question # 22 of 30 (Start time: 11:44:17 AM, 27 June 2021)

The O-notation is used to state only the asymptotic _____ bounds.

Select the correct option

☐	Lower
☐	Both lower & upper
☒	Upper
☐	Two





CS502:Grand Quiz

Question # 23 of 30 (Start time: 11:44:44 AM, 27 June 2021)

For average-case time analysis of quick sort algorithm, pivot selection is on average basis from _____.

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☐ | values greater than 5 |
| ☐ | all values except even |
| ☒ | all possible random values |
| ☐ | half of the input values |





CS502:Grand Quiz

Question # 24 of 30 (Start time: 11:45:12 AM, 27 June 2021)

Upper bound requires that there exist positive constants c_2 and n_0 such that $f(n) \leq c_2 n$ for all $n \geq n_0$

Select the correct option

☐	Less than
☐	Equal or Greater than
☒	Equal to or Less than
☐	Greater than





Question # 25 of 30 (Start time: 11:45:41 AM, 27 June 2021)

Number of _____ of the pseudo code are counted to measure the running time.

Select the correct option

☐	Inputs
☒	Steps
☐	Pages
☐	Outputs

Click to Save Answer 8





CS502:Grand Quiz

Question # 26 of 30 (Start time: 11:46:07 AM, 27 June 2021)

In Dynamic Programming, our approach is to _____.

Select the correct option

- | | |
|----------------------------------|--|
| ☒ | Build the solution in a bottom-up fashion |
| ☐ | Input several sub-problems simultaneously |
| ☐ | Develop the solution in a top-down fashion |
| ☐ | Express the problem non-recursively |





CS502:Grand Quiz

Question # 27 of 30 (Start time: 11:46:35 AM, 27 June 2021)

In the statement "output P[i].x, P[i].y", the number of times elements of P are accessed is _____

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | 3 |
| ☒ | 2 |
| ☐ | 1 |
| ☐ | 4 |





CS502:Grand Quiz

Question # 28 of 30 (Start time: 11:47:15 AM, 27 June 2021)

Sorting Algorithms having O _____ running time are considered to be slow ones.

Select the correct option

- | | |
|----------------------------------|--------------|
| ☐ | $(\log(n))$ |
| ☐ | $(n\log(n))$ |
| ☐ | (n) |
| ☒ | (n^2) |





CS502:Grand Quiz

Question # 29 of 30 (Start time: 11:47:42 AM, 27 June 2021)

There are _____ entries in the Edit Distance Matrix.

Select the correct option

- | | |
|----------------------------------|-------------------|
| ☐ | $\Theta(n + 100)$ |
| ☐ | $\Theta(n+2)$ |
| ☐ | $\Theta(n)$ |
| ☒ | $\Theta(n^2)$ |





CS502:Grand Quiz

Question # 30 of 30 (Start time: 11:48:09 AM, 27 June 2021)

In Heap Sort algorithm, if heap property is violated:

Select the correct option

- | | |
|----------------------------------|-------------------------------------|
| ☐ | We call Build heap procedure |
| ☒ | We call Heapify procedure |
| ☐ | We ignore |
| ☐ | Heap property can never be violated |





CS502:Grand Quiz

Question # 3 of 30 (Start time: 12:03:39 PM, 27 June 2021)

Recurrence can be described in terms of _____.

Select the correct option

☐	Array
☐	Linear
☐	Graph
☒	Tree

Click to Save





Question # 4 of 30 (Start time: 12:04:10 PM, 27 June 2021)

In his book _____, Leonardo Pisano addressed the Fibonacci sequence as well as a variety of other problems.

Select the correct option

- | | |
|----------------------------------|-------------|
| ☐ | Liber fib |
| ☒ | Liber abaci |
| ☐ | Fibonacci |
| ☐ | Fib abaci |

Click to Save Answer &





Question # 5 of 30 (Start time: 12:04:40 PM, 27 June 2021)

In asymptotical analysis of $n(5 + 2) - 3$, as n becomes large, the dominant (fastest growing) term is some constant times _____

Select the correct option

☒	n
☐	$n*n$
☐	n_1
☐	$n+1$

Click to Save Answer





CS502:Grand Quiz

Question # 6 of 30 (Start time: 12:05:07 PM, 27 June 2021)

The use of the temporary array in merge sort is a (n) _____ of merge sort.

Select the correct option

☒	shortcoming
☐	blessing
☐	strength
☐	advantage

Click to Save





Question # 7 of 30 (Start time: 12:05:30 PM, 27 June 2021)

In order to analyze algorithm, a Mathematical Model of computation is used known as

Select the correct option

- | | |
|----------------------------------|-----------------------|
| ☐ | Read Only Machine |
| ☐ | Read drive Machine |
| ☐ | None of the above |
| ☒ | Random Access Machine |

Click to Save Answer & Move





Question # 8 of 30 (Start time: 12:05:58 PM, 27 June 2021)

Point p is said to dominated by point q if:

Select the correct option

- | | |
|----------------------------------|-----------------------------------|
| ☐ | $p.y \leq q.y$ |
| ☐ | $p.x \leq q.x$ |
| ☐ | $p.x \leq q.x$ or $p.y \leq q.y$ |
| ☒ | $p.x \leq q.x$ and $p.y \leq q.y$ |

Click to Save Answer





Question # 9 of 30 (Start time: 12:06:31 PM, 27 June 2021)

Total Marks

You are given a knapsack that can carry a maximum weight of 60. There are 4 items with weights {20, 30, 40, 70} and values {90, 80, 100, 250}. What is the maximum value of the items you can carry using the knapsack?

Select the correct option

[Reload Math Equation](#)

☐	520
☐	270
☒	190
☐	170

[Click to Save Answer & Move to Next Question](#)

Question # 10 of 30 (Start time: 12:07:04 PM, 27 June 2021)

Total Marks: 1

You are given a knapsack that can carry a maximum weight of 90. There are 4 items with weights {50, 40, 60, 70} and values {90, 80, 100, 250}. What is the maximum value of the items you can carry using the knapsack?

Select the correct option

[Reload Math Equations](#)

☐	180
☐	350
☐	190
☒	170

[Click to Save Answer & Move to Next Question](#)



Question # 11 of 30 (Start time: 12:07:33 PM, 27 June 2021)

In Brute Force solution of Knapsack Problem, there are _____ possible combinations for n items.

Select the correct option

☐	n
☒	2^n (2 raise to the power n)
☐	2
☐	3^n (3 raise to the power n)





Question # 13 of 30 (Start time: 12:08:46 PM, 27 June 2021)

_____ is the process of avoiding unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later.

Select the correct option

- | | |
|----------------------------------|-------------|
| ☐ | Loop |
| ☐ | Function |
| ☐ | Recursion |
| ☒ | Memoization |

[Click to Save Answer & Move to Next](#)

Question # 14 of 30 (Start time: 12:09:14 PM, 27 June 2021)

Suppose we have 4 matrices A, B, C, D. What is correct expansion of $m[1,2]$ in chain matrix multiplication?

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | $m[1, 2] = m[1, 1] + m[2, 2] + p_0 \cdot p_1 \cdot p_3$ |
| ☐ | $m[1, 2] = m[1, 2] + m[2, 2] + p_0 \cdot p_1 \cdot p_2$ |
| ☐ | $m[1, 2] = m[1, 1] + m[1, 2] + p_0 \cdot p_1 \cdot p_2$ |
| ☒ | $m[1, 2] = m[1, 1] + m[2, 2] + p_0 \cdot p_1 \cdot p_2$ |

[Click to Save Answer & Move to Next](#)



Question # 15 of 30 (Start time: 12:09:52 PM, 27 June 2021)

In Sorting the key value or attribute _____ from an ordered domain.

Select the correct option

☐	Occasionally
☐	May be
☒	Must be
☐	Not always

Click to Save Answer





Question # 16 of 30 (Start time: 12:10:24 PM, 27 June 2021)

Radix sort is a _____ integer sorting algorithm.

Select the correct option

☐	Unstable
☐	Comparative
☐	In-Place
☒	Non-comparative

Click to Save Ans





CS502:Grand Quiz

Question # 17 of 30 (Start time: 12:10:51 PM, 27 June 2021)

In plane sweep approach, a vertical line is swept across the 2d-plane from_____.

Select the correct option

☐	Top to Bottom
☒	Left to Right
☐	Bottom to Top
☐	Right to Left



Question # 18 of 30 (Start time: 12:11:25 PM, 27 June 2021)

If there are $\Theta(n^2)$ entries in edit distance matrix then each entry $E(i, j)$ takes _____ time to compute.

Select the correct option

☐	$\Theta(n)$
☒	$\Theta(1)$
☐	$\Theta(n^2)$
☐	$\Theta(n \log n)$





Question # 19 of 30 (Start time: 12:11:55 PM, 27 June 2021)

_____ of reference is an important fact of current processor technology.

Select the correct option

☐	Formality
☐	Assigning
☐	Defining
☒	Locality

Click to Save Answer & Mo





CS502:Grand Quiz

Question # 20 of 30 (Start time: 12:12:22 PM, 27 June 2021)

Two functions $4n^2$ and $(8n^2 + 2n - 3)$ are _____.

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☐ | hard to calculate |
| ☐ | related to power functions |
| ☒ | asymptotically equivalent |
| ☐ | not equivalent |

Click to Save





CS502:Grand Quiz

Question # 21 of 30 (Start time: 12:12:52 PM, 27 June 2021)

For $n > 1$, _____ divides into two halves, sorts the two and then combine them together.

Select the correct option

☐	Quick Sort
☒	MergeSort
☐	Insertion Sort
☐	Heap Sort

Click to Save





Question # 22 of 30 (Start time: 12:13:18 PM, 27 June 2021)

While applying the sieve technique, _____ subarray will contain all elements that are less than pivot element x.

Select the correct option



- | | |
|----------------------------------|-------------|
| ☐ | A[1....n] |
| ☐ | A[q] |
| ☒ | A[1....q-1] |
| ☐ | A[q+1....n] |



Question # 23 of 30 (Start time: 12:14:05 PM, 27 June 2021)

In max heap (for Heap Sort algorithm), when every time maximum element is removed from top we replace it with _____ leaf in the tree.

Select the correct option

☒	Last
☐	First
☐	Any
☐	Second last

Click to Save Answer & Move





CS502:Grand Quiz

Question # 24 of 30 (Start time: 12:14:38 PM, 27 June 2021)

For small values of n , _____ algorithm is fast enough.

Select the correct option

☐	quick sort
☐	merge sort
☒	Any
☐	heap sort





Question # 25 of 30 (Start time: 12:15:05 PM, 27 June 2021)

While applying the sieve technique, _____ subarray will contain all elements that are greater than pivot element x.

Select the correct option

- | | |
|----------------------------------|------------------|
| ☐ | $A[1 \dots q-1]$ |
| ☒ | $A[q+1 \dots n]$ |
| ☐ | $A[1 \dots n]$ |
| ☐ | $A[q]$ |

Click to Save Answ





Question # 26 of 30 (Start time: 12:15:40 PM, 27 June 2021)

In _____ sorting algorithm, we just need to swap positions of data during the Partitioning function.

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | Counting sort |
| ☐ | Merge sort |
| ☐ | Radix sort |
| ☒ | Quick sort |

Click to Save Answer &





CS502:Grand Quiz

Question # 27 of 30 (Start time: 12:16:07 PM, 27 June 2021)

The worst case running time of quick sort algorithm _____.

Select the correct option

☐	Is linear
☐	Cannot be quadratic
☐	Is always Exponential
☒	Is quadratic

Click to Save





CS502:Grand Quiz

Question # 28 of 30 (Start time: 12:16:39 PM, 27 June 2021)

An in-place sorting algorithm is one that _____ use(s) additional array for storage.

Select the correct option

☐	Sometime
☒	Does not
☐	Permanently
☐	Always





Question # 29 of 30 (Start time: 12:17:03 PM, 27 June 2021)

In the following code the statement "cout<<i;" executes ----- times.
for (int i = 1; i<=n ;i++)
cout<<i;

Select the correct option

☐	Infinite times
☐	Zero times
☐	n+5 times
☒	n times





Question # 30 of 30 (Start time: 12:17:33 PM, 27 June 2021)

One of the limitation in 0/1 knapsack is that an item can either be _____ in the bag or not.

Select the correct option

☐	Move
☐	Use
☐	Store
☒	Put

Click to Save Answer





Question # 1 of 30 (Start time: 12:20:08 PM, 27 June 2021)

In Running Time Analysis, we are also concerned about the _____ required by the algorithm.

Select the correct option

- | | |
|----------------------------------|-----------------------|
| ☐ | Space |
| ☐ | code |
| ☒ | Both space and memory |
| ☐ | Memory |

Click to Save Answer





CS502:Grand Quiz

Question # 2 of 30 (Start time: 12:21:02 PM, 27 June 2021)

The only way to convert an empty string into a string of j characters is by doing j insertions, represented as _____.

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | $E(1, j) = j$ |
| ☐ | $E(i, 0) = i$ |
| ☒ | $E(0, j) = j$ |
| ☐ | $E(i, j) = i$ |

Click to Save





Question # 3 of 30 (Start time: 12:22:05 PM, 27 June 2021)

In Heap Sort algorithm, to remove the maximum element every time,_____.

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | Nothing happens |
| ☐ | Heap Sort algorithm terminates without result |
| ☐ | We call Build-Heap procedure |
| ☒ | We call heapify procedure |

Click to Save Answer & Move to Next Question





CS502:Grand Quiz

Question # 4 of 30 (Start time: 12:22:45 PM, 27 June 2021)

The formula for calculating the Catalan number is _____.

Select the correct option

- | | |
|----------------------------------|--------------------------------------|
| ☐ | $C(n) = \frac{1}{n-1} \binom{n}{2n}$ |
| ☐ | $C(n) = \frac{1}{n-1} \binom{2n}{n}$ |
| ☒ | $C(n) = \frac{1}{n+1} \binom{2n}{n}$ |
| ☐ | $C(n) = \frac{1}{n+1} \binom{n}{2n}$ |



Question # 5 of 30 (Start time: 12:23:14 PM, 27 June 2021)

For average-case time analysis of quick sort algorithm, pivot selection is on average basis from _____.

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☐ | all values except even |
| ☐ | half of the input values |
| ☐ | values greater than 5 |
| ☒ | all possible random values |

Click to Save Answer &





CS502:Grand Quiz

Question # 6 of 30 (Start time: 12:23:42 PM, 27 June 2021)

Chain matrix multiplication problem can be solved through _____ strategy.

Select the correct option

☒	dynamic programming
☐	greedy
☐	divide and conquer
☐	Implementing arrays





CS502:Grand Quiz

Question # 7 of 30 (Start time: 12:24:02 PM, 27 June 2021)

In chain matrix multiplication, table is filled _____ to find the multiplication of matrix.

Select the correct option

☒	diagonally
☐	column wise
☐	row wise
☐	bottom-to-up

Click





Question # 8 of 30 (Start time: 12:24:27 PM, 27 June 2021)

PUSH and POP terminologies are used in _____.

Select the correct option

- | | |
|----------------------------------|--------------------|
| ☐ | circular queue |
| ☐ | double-ended queue |
| ☐ | queue |
| ☒ | stack |

Click to Save Answer





Question # 9 of 30 (Start time: 12:24:52 PM, 27 June 2021)

Total Marks: 1

You are given a knapsack that can carry a maximum weight of 90. There are 4 items with weights {50, 40, 60, 70} and values {90, 80, 100, 250}. What is the maximum value of the items you can carry using the knapsack?

Select the correct option

Reload Math Equations

☐	190
☐	350
☐	180
☒	170

Click to Save Answer & Move to Next Question





CS502:Grand Quiz

Question # 10 of 30 (Start time: 12:25:19 PM, 27 June 2021)

What is the value of k in the given equation?

$$m[2, 5] = m[2, 3] + m[4, 5] + p1 \cdot p3 \cdot p5$$

Select the correct option

☒	3
☐	2
☐	5
☐	4



CS502:Grand Quiz

Question # 11 of 30 (Start time: 12:25:43 PM, 27 June 2021)

Quick sort Algorithm is required a lot of comparisons in the _____ condition.

Select the correct option

☒	Worse case
☐	Average case
☐	Best case
☐	Best and Average case

CS502:Grand Quiz

Quiz Start Time: 12:20 PM, 27

Question # 12 of 30 (Start time: 12:26:08 PM, 27 June 2021)

To

If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions

Select the correct option

☒	3x3
☐	2x2
☐	3x2
☐	2x3





Question # 13 of 30 (Start time: 12:26:30 PM, 27 June 2021)

Which type of operations does RAM perform?

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☐ | Access element of array |
| ☐ | Comparison or Boolean |
| ☒ | All of above |
| ☐ | Assign value to a variable |

Click to Save Answer





CS502:Grand Quiz

Question # 14 of 30 (Start time: 12:26:59 PM, 27 June 2021)

Knapsack word originate from _____ language.

Select the correct option

☒	German
☐	English
☐	Norwegian
☐	French

Click





CS502:Grand Quiz

Question # 15 of 30 (Start time: 12:27:24 PM, 27 June 2021)

_____ items are not allowed in 0/1 Knapsack problem.

Select the correct option

☐	0
☒	Fractional
☐	0/1
☐	1

Click to Save





Question # 16 of 30 (Start time: 12:27:51 PM, 27 June 2021)

In _____, Leonardo of Pisa, also called Fibonacci, published a book.

Select the correct option

- | | |
|----------------------------------|------|
| ☒ | 1202 |
| ☐ | 1400 |
| ☐ | 1102 |
| ☐ | 1304 |

Click to Save Answer





Question # 17 of 30 (Start time: 12:28:42 PM, 27 June 2021)

In his book _____, Leonardo Pisano addressed the Fibonacci sequence as well as a variety of other problems.

Select the correct option

Reload M

- | | |
|----------------------------------|-------------|
| ☐ | Fib abaci |
| ☐ | Fibonacci |
| ☒ | Liber abaci |
| ☐ | Liber fib |

Click to Save Answer & Move to Next





CS502:Grand Quiz

Question # 18 of 30 (Start time: 12:29:17 PM, 27 June 2021)

There are no hard formal rules to the syntax of the _____ code.

Select the correct option

☐	Basic
☒	Pseudo
☐	Assembly
☐	Programming





CS502:Grand Quiz

Question # 19 of 30 (Start time: 12:29:53 PM, 27 June 2021)

In max-heap, largest element is stored at root node. Where is the smallest element stored?

Select the correct option

- | | |
|----------------------------------|-------------|
| ☐ | Middle Node |
| ☐ | Right Node |
| ☐ | Left Node |
| ☒ | Leaf Node |



CS502:Grand Quiz

Question # 20 of 30 (Start time: 12:30:23 PM, 27 June 2021)

If the indices passed to merge sort algorithm are _____, then this means that there is only one element to sort.

Select the correct option

☐	Small
☐	Large
☒	Equal
☐	Not Equal





Question # 21 of 30 (Start time: 12:30:54 PM, 27 June 2021)

If we have an equation $8n^2 + 7f \cdot n + 5f + 6$ then n is large, _____ term will be much larger than the n term and will dominate the running time.

Select the correct option

☐	$g(n)^2$
☐	$f(n)$
☐	$f(g(n))$
☒	n^2

[Click to Save Answer & Move to Next Question](#)



Question # 22 of 30 (Start time: 12:31:24 PM, 27 June 2021)

If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.

Select the correct option

☒	Cheap
☐	Slow
☐	Expensive
☐	Fast

Click to Save Answer & Move to





CS502:Grand Quiz

Question # 23 of 30 (Start time: 12:31:58 PM, 27 June 2021)

In the following loop statement:

for i ← 1 to n

For loop executes _____ times.

Select the correct option

☒	n
☐	It is not correct statement
☐	n^2
☐	log n





CS502:Grand Quiz

Question # 24 of 30 (Start time: 12:32:29 PM, 27 June 2021)

The asymptotic growth of $n(n+1)/2$ is:

Select the correct option

☐	$O(n \log n)$
☐	$O(n+2)$
☒	$O(n^2)$
☐	$O(n)$

Click to





CS502:Grand Quiz

Question # 25 of 30 (Start time: 12:33:12 PM, 27 June 2021)

_____ programming is essentially recursion without repetition.

Select the correct option

☒	Dynamic
☐	Array
☐	$N \log n$
☐	Fast

Click





Question # 26 of 30 (Start time: 12:33:34 PM, 27 June 2021)

Consider two matrices A and B, where A is 10×30 and B is 30×40 . What is correct number of multiplications require to multiply A and B?

Select the correct option

Relo

- | | |
|----------------------------------|--------------------------|
| ☐ | 30×30 |
| ☐ | 10×30 |
| ☐ | 30×40 |
| ☒ | $10 \times 30 \times 40$ |





Question # 27 of 30 (Start time: 12:34:04 PM, 27 June 2021)

While applying the Sieve technique to selection sort, how to choose a pivot element.

Select the correct option

- | | |
|----------------------------------|--------------|
| ☐ | Linear |
| ☐ | Through mean |
| ☒ | Randomly |
| ☐ | Sequentially |

Click to Save Ans





Question # 28 of 30 (Start time: 12:34:41 PM, 27 June 2021)

In chain matrix multiplication, the order of the matrices_____.

Select the correct option

☒	can not be changed
☐	is equal
☐	can be changed
☐	is reverse

Click to Save A





CS502:Grand Quiz

Question # 29 of 30 (**Start time: 12:35:05 PM, 27 June 2021**)

Following is not the application of Edit Distance problem.

Select the correct option

☐	Speech recognition
☐	Spelling Correction
☐	Computational Molecular Biology
☒	Ascending Sort

Click to





CS502:Grand Quiz

Question # 30 of 30 (Start time: 12:35:35 PM, 27 June 2021)

In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object 'i', then we gain _____.

Select the correct option

☐	None of the given option
☐	V (Total Value of all items)
☐	W (Total Weight of Knapsack)
☒	v_i (Value of object i)





Question # 1 of 30 (Start time: 01:17:48 PM, 27 June 2021)

In Random Access Machine, Instructions are executed

Select the correct option

☐	Three at a time
☐	Five at a time
☒	One by one
☐	Multiple in parallel





CS502:Grand Quiz

Question # 2 of 30 (Start time: 01:18:33 PM, 27 June 2021)

In Heap Sort algorithm, the first step is to _____.

Select the correct option

☐	Sort the array in descending order
☒	Call Build-Heap procedure
☐	Find the number of input elements
☐	Call Heapify procedure



Question # 3 of 30 (Start time: 01:19:03 PM, 27 June 2021)

Quick sort is based on _____ strategy.

Select the correct option

☐	Greedy approach
☒	Divide-and-Conquer
☐	Dynamic programming
☐	Graph Theory

Click to Save Answer





CS502:Grand Quiz

Question # 4 of 30 (Start time: 01:19:27 PM, 27 June 2021)

In bin sort, each bin can be replaced by a _____ in case of duplication.

Select the correct option

☐	Stack
☐	bin
☒	linked list
☐	Queue

Click





Question # 5 of 30 (Start time: 01:19:52 PM, 27 June 2021)

In Selection problem, the Sieve technique works in _____

Select the correct option

☒	Phases
☐	Constant time
☐	One complete go
☐	Non-recursive manner

Click to Save Answer &





CS502:Grand Quiz

Question # 6 of 30 (Start time: 01:20:16 PM, 27 June 2021)

Quicksort is a/an _____ and _____ sorting algorithm.

Select the correct option

- | | |
|----------------------------------|------------------------------|
| ☐ | Not in-place, not stable one |
| ☒ | In-place, not stable one |
| ☐ | Not in-place, stable one |
| ☐ | In-place, stable one |





Question # 7 of 30 (Start time: 01:20:46 PM, 27 June 2021)

In _____ we have to find rank of an element from given input.

Select the correct option

☐	Plane Sweep algorithm
☒	Selection problem
☐	Brute force technique
☐	Merge sort algorithm

Click to Save Answ





Question # 8 of 30 (Start time: 01:21:20 PM, 27 June 2021)

_____ is a method of solving a problem in which we check all possible solutions to the problem to find the solution we need.

Select the correct option

☒	Brute-Force Algorithm
☐	Sorting Algorithm
☐	Plane-Sweep Algorithm
☐	Greedy approach





Question # 9 of 30 (Start time: 01:21:43 PM, 27 June 2021)

For average-case time analysis of quick sort algorithm, pivot selection is on average basis from _____.

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☐ | values greater than 5 |
| ☒ | all possible random values |
| ☐ | half of the input values |
| ☐ | all values except even |

Click to Save Answer & Move to





Question # 10 of 30 (Start time: 01:22:08 PM, 27 June 2021)

We can use the optimal substructure property to devise a _____ formulation of the edit distance problem.

Select the correct option

- | | |
|----------------------------------|-----------|
| ☐ | Optimum |
| ☐ | Iterative |
| ☒ | Recursive |
| ☐ | Selective |

Click to Save Answer &





Question # 11 of 30 (Start time: 01:22:37 PM, 27 June 2021)

Counting sort assumes that the numbers to be sorted are in the range _____.

Select the correct option

- | | |
|----------------------------------|-------------------------|
| ☐ | k to n where k is small |
| ☐ | k to n where n is small |
| ☐ | k to n where n is large |
| ☒ | 1 to k where k is small |

Click to Save Answer & Move to Next Question





Question # 12 of 30 (Start time: 01:23:07 PM, 27 June 2021)

In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object 'i', then we gain _____.

Select the correct option

☐	None of the given option
☐	V (Total Value of all items)
☐	W (Total Weight of Knapsack)
☒	v_i (Value of object i)

Click to Save Answer





CS502:Grand Quiz

Question # 13 of 30 (Start time: 01:23:36 PM, 27 June 2021)

Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

Select the correct option

☒	All
☐	Most
☐	Some
☐	Two

Click t





Question # 14 of 30 (Start time: 01:23:59 PM, 27 June 2021)

Approach of solving geometric problems by sweeping a line across the plane is called _____sweep.

Select the correct option

☐	Box
☐	Cube
☒	Plane
☐	Line

Click to Save Answer & Move to Next Question





Question # 15 of 30 (Start time: 01:24:46 PM, 27 June 2021)

When matrix A of 5 x 3 is multiplied with matrix B of 3 x 4 then the number of multiplications required will be _____.

Select the correct option

- | | |
|----------------------------------|----|
| ☒ | 60 |
| ☐ | 15 |
| ☐ | 12 |
| ☐ | 36 |

Click to Save Answer





Question # 16 of 30 (Start time: 01:25:09 PM, 27 June 2021)

We can use the _____ property to devise a recursive formulation of the edit distance problem.

Select the correct option

☐	small substructure
☒	optimal substructure
☐	real
☐	algorithmic

Click to Submit





CS502:Grand Quiz

Question # 17 of 30 (**Start time: 01:25:41 PM, 27 June 2021**)

Quick sort is based on _____ strategy.

Select the correct option

- | | |
|----------------------------------|---------------------|
| ☐ | Graph Theory |
| ☐ | Dynamic programming |
| ☒ | Divide-and-Conquer |
| ☐ | Greedy approach |

Click t





CS502:Grand Quiz

Question # 18 of 30 (Start time: 01:26:09 PM, 27 June 2021)

Asymptotic growth of $8n^2 + 2n - 3$ is :

Select the correct option

- | | |
|----------------------------------|---------------------|
| ☐ | $\Theta(8n^2)$ |
| ☐ | $\Theta(n^2 + n)$ |
| ☒ | $\Theta(n^2)$ |
| ☐ | $\Theta(8n^2 + 2n)$ |



Question # 19 of 30 (Start time: 01:26:41 PM, 27 June 2021)

The optimal solution to a problem is a combination of optimal solutions to its subproblems. This is known as _____.

Select the correct option



- | | |
|----------------------------------|--------------------------|
| ☒ | Principle of optimality |
| ☐ | Principle of locality |
| ☐ | Principle of computation |
| ☐ | Principle of feasibility |

Click to Save Answer & Move to Next Question





Question # 20 of 30 (Start time: 01:27:05 PM, 27 June 2021)

The sieve technique works where we have to find _____ item(s) from a large input.

Select the correct option

☐	Two
☐	Three
☐	Similar
☒	Single

Click to Save Answer





CS502:Grand Quiz

Question # 21 of 30 (Start time: 01:27:32 PM, 27 June 2021)

In Heap Sort algorithm, if heap property is violated:

Select the correct option

- | | |
|----------------------------------|-------------------------------------|
| ☒ | We call Heapify procedure |
| ☐ | We call Build heap procedure |
| ☐ | Heap property can never be violated |
| ☐ | We ignore |

CS502:Grand Quiz

Quiz Start Time: 01:17 PM, 27 Ju

Question # 22 of 30 (Start time: 01:27:55 PM, 27 June 2021)

Total

In Ram, each instruction involves performing some basic operation on how many values in the machines memory?

Select the correct option

[Reload Math Equ](#)

- | | |
|----------------------------------|-------|
| ☐ | Three |
| ☒ | Two |
| ☐ | One |
| ☐ | Four |

[Click to Save Answer & Move to Next Que](#)



Question # 23 of 30 (Start time: 01:28:22 PM, 27 June 2021)

In max-heap, largest element is stored at root node. Where is the smallest element stored?

Select the correct option

☐	Middle Node
☒	Leaf Node
☐	Left Node
☐	Right Node





Question # 24 of 30 (Start time: 01:28:59 PM, 27 June 2021)

In Selection algorithm, we assume pivot selection takes theta _____ running time.

Select the correct option

- | | |
|----------------------------------|-----------|
| ☐ | $\log(n)$ |
| ☐ | n^2 |
| ☐ | n^3 |
| ☒ | n |

[Click to Save Answer](#)

Question # 25 of 30 (Start time: 01:29:36 PM, 27 June 2021)

Total Marks: 1

In plane sweep approach, a vertical line is swept across the 2d-plane and _____ structure is used for holding the maximal points lying to the left of the sweep line.

Select the correct option

- | | |
|----------------------------------|-------|
| ☐ | Tree |
| ☐ | Queue |
| ☒ | Stack |
| ☐ | Array |





Question # 26 of 30 (Start time: 01:30:05 PM, 27 June 2021)

Consider three matrices X, Y, Z of dimensions 1×2 , 2×3 , 3×4 respectively. The number of multiplications of $X(YZ)$ is:

Select the correct option

- | | |
|----------------------------------|----|
| ☐ | 32 |
| ☐ | 16 |
| ☐ | 26 |
| ☒ | 32 |





CS502:Grand Quiz

Question # 27 of 30 (Start time: 01:30:36 PM, 27 June 2021)

Selection sort is a _____ sorting algorithm.

Select the correct option

☐	in-partition
☒	In-place
☐	Stable
☐	Not In-Place





Question # 28 of 30 (Start time: 01:31:00 PM, 27 June 2021)

If matrix A of dimension $p \times q$ is multiply with matrix B of dimension $q \times r$, then the dimension of resultant matrix is:

Select the correct option

- | | |
|----------------------------------|--------------|
| ☐ | $r \times p$ |
| ☐ | $p \times q$ |
| ☐ | $q \times r$ |
| ☒ | $p \times r$ |

[Click to Save Answer & Move](#)



CS502:Grand Quiz

Question # 29 of 30 (Start time: 01:31:33 PM, 27 June 2021)

In the analysis of plane-sweep algorithm, sorting takes _____.

Select the correct option

- | | |
|----------------------------------|--------------------|
| ☐ | $\theta(n^2)$ |
| ☐ | $\theta(n)$ |
| ☐ | $\theta(\log n)$ |
| ☒ | $\theta(n \log n)$ |

Click





CS502:Grand Quiz

Question # 30 of 30 (Start time: 01:32:08 PM, 27 June 2021)

RAM is idealized machine with _____ large random-access memory.

Select the correct option

☐	Small
☐	Finite
☒	Infinite
☐	Average





CS502:Grand Quiz

Question # 1 of 30 (Start time: 01:35:51 PM, 27 June 2021)

_____ is always based on divide and conquer strategy.

Select the correct option

☐	Bubble sort
☐	Selection sort
☒	Quick sort
☐	Pigeon sort





Question # 2 of 30 (Start time: 01:36:36 PM, 27 June 2021)

Which of the following is calculated with Big Omega notation?

Select the correct option

- | | |
|----------------------------------|-----------------------------|
| ☐ | Both upper and lower bounds |
| ☒ | Lower bounds |
| ☐ | Upper bounds |
| ☐ | Medium bounds |





CS502:Grand Quiz

Question # 3 of 30 (Start time: 01:37:00 PM, 27 June 2021)

Following is not the application of Edit Distance problem.

Select the correct option

☐	Spelling Correction
☐	Speech recognition
☐	Computational Molecular Biology
☒	Ascending Sort



CS502:Grand Quiz

Question # 4 of 30 (Start time: 01:37:25 PM, 27 June 2021)

While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

Select the correct option

☒	All of the above
☐	$x = y$
☐	$x < y$
☐	$x > y$



Question # 5 of 30 (Start time: 01:37:56 PM, 27 June 2021)

Suppose we have 5 matrices A, B, C, D and E. What is the correct expansion of $m[3,4]$ in chain matrix multiplication?

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | $m[3, 4] = m[3, 4] + m[4, 4] + p_2 \cdot p_3 \cdot p_4$ |
| ☒ | $m[3, 4] = m[3, 3] + m[4, 4] + p_2 \cdot p_3 \cdot p_4$ |
| ☐ | $m[3, 4] = m[3, 3] + m[4, 4] + p_0 \cdot p_3 \cdot p_4$ |
| ☐ | $m[3, 4] = m[3, 3] + m[3, 4] + p_2 \cdot p_3 \cdot p_4$ |

[Click to Save Answer](#)



CS502:Grand Quiz

Question # 6 of 30 (Start time: 01:38:22 PM, 27 June 2021)

If the time complexity of an algorithm is given by $O(1)$, then its time complexity would be:

Select the correct option

☐	Polynomial
☐	Exponential
☒	Constant
☐	Average

Click to





CS502:Grand Quiz

Question # 7 of 30 (Start time: 01:38:58 PM, 27 June 2021)

If input "n" is odd, then median will be _____

Select the correct option

☐	$n/2$
☐	$n + 2$
☒	$(n+1)/2$
☐	$(n-1)/2$





Question # 8 of 30 (Start time: 01:39:29 PM, 27 June 2021)

Which process is used for avoiding unnecessary repetitions and looking them up again if we need them later.

Select the correct option

☐	Recursion
☐	Greedy Approach
☒	Memoization
☐	Divide and conquer





Question # 9 of 30 (Start time: 01:39:58 PM, 27 June 2021)

In generating Fibonacci Sequence, we can avoid unnecessary repetitions by _____ process.

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | Randomization |
| ☐ | Memorization |
| ☐ | Tokenization |
| ☒ | Memoization |

Click to Save Answer & M





CS502:Grand Quiz

Question # 10 of 30 (Start time: 01:40:24 PM, 27 June 2021)

In Fibonacci Sequence, each term is calculated by _____ previous _____ terms.

Select the correct option

- | | |
|----------------------------------|------------------|
| ☐ | Multiplying, Two |
| ☒ | Adding, Two |
| ☐ | Subtracting, Two |
| ☐ | Adding, Three |

Click





CS502:Grand Quiz

Question # 11 of 30 (Start time: 01:40:54 PM, 27 June 2021)

Catalan numbers are related the number of different _____ on 'n' nodes.

Select the correct option

☒	binary trees
☐	linked lists
☐	functions
☐	arrays



CS502:Grand Quiz

Question # 12 of 30 (Start time: 01:41:18 PM, 27 June 2021)

Quick sort is based on _____ strategy.

Select the correct option

- | | |
|----------------------------------|---------------------|
| ☐ | Dynamic programming |
| ☒ | Divide-and-Conquer |
| ☐ | Greedy approach |
| ☐ | Graph Theory |

CS502:Grand Quiz

Quiz Start Time: 01:35 PM, 27 June 2021

Question # 13 of 30 (Start time: 01:41:45 PM, 27 June 2021)

Total Marks: 1

If matrix A of dimension 2×4 is multiplied with matrix B of dimension 4×3 , then the dimension of resultant matrix will be _____.

Select the correct option

- | | |
|----------------------------------|--------------|
| ☐ | 4×3 |
| ☐ | 2×4 |
| ☐ | 3×4 |
| ☒ | 2×3 |

[Click to Save Answer & Move to Next Question](#)



CS502:Grand Quiz

Question # 14 of 30 (Start time: 01:42:30 PM, 27 June 2021)

In Heap Sort algorithm, we call Build-heap procedure _____.

Select the correct option

☐	Thrice
☐	Twice
☐	As many times as we need
☒	Only once





CS502:Grand Quiz

Question # 15 of 30 (Start time: 01:42:57 PM, 27 June 2021)

The main purpose of mathematical analysis is measuring the _____ required by the algorithm.

Select the correct option

☐	Execution time
☐	Inputs & outputs
☒	Execution time and memory
☐	Space





CS502:Grand Quiz

Question # 16 of 30 (Start time: 01:43:26 PM, 27 June 2021)

_____ algorithm based on Dynamic Programming strategy.

Select the correct option

☐	Heap Sort
☐	Quick Sort
☐	Binary Tree
☒	Edit distance





CS502:Grand Quiz

Question # 17 of 30 (Start time: 01:44:08 PM, 27 June 2021)

_____ belongs to Dynamic programming.

Select the correct option

☐	Merge sort
☒	Edit distance
☐	Divide and conquer
☐	Heap sort





CS502:Grand Quiz

Question # 18 of 30 (Start time: 01:44:36 PM, 27 June 2021)

If there are duplicates in _____ sort, then each bin can be replaced by a linked list.

Select the correct option

- | | |
|----------------------------------|--------|
| ☒ | Bucket |
| ☐ | Merge |
| ☐ | Heap |
| ☐ | Quick |





CS502:Grand Quiz

Question # 19 of 30 (Start time: 01:45:03 PM, 27 June 2021)

Result of asymptotical analysis of $n(n-3)$ and $4n*n$ is that _____.

Select the correct option

☐	$n(n-1)$ is asymptotically Less
☐	$n(n-1)$ is asymptotically Greater
☒	Both are asymptotically Equivalent
☐	Both are asymptotically Not equivalent





Question # 20 of 30 (Start time: 01:45:31 PM, 27 June 2021)

If there are $\Theta(n^2)$ entries in edit distance matrix then the total running time is:

Select the correct option

- | | |
|----------------------------------|--------------------|
| ☐ | $\Theta(1)$ |
| ☒ | $\Theta(n^2)$ |
| ☐ | $\Theta(n)$ |
| ☐ | $\Theta(n \log n)$ |

Click to Save Answer





Question # 21 of 30 (Start time: 01:46:00 PM, 27 June 2021)

In chain matrix multiplication, table is filled _____ to find the multiplication of matrix.

Select the correct option

☐	column wise
☐	row wise
☐	bottom-to-up
☒	diagonally

Click to Save Answer





CS502:Grand Quiz

Question # 22 of 30 (Start time: 01:46:31 PM, 27 June 2021)

In the following loop statement:

for i ← 1 to n

For loop executes _____ times.

Select the correct option

☐	log n
☐	n^2
☒	n
☐	It is not correct statement





Question # 23 of 30 (Start time: 01:46:57 PM, 27 June 2021)

In chain matrix multiplication, solutions of the sub-problems are stored in a _____.

Select the correct option

☐	Tree
☐	Array
☒	Table
☐	Linked list





Question # 24 of 30 (Start time: 01:47:24 PM, 27 June 2021)

We can use the _____ property to devise a recursive formulation of the edit distance problem.

Select the correct option

- | | |
|----------------------------------|----------------------|
| ☒ | optimal substructure |
| ☐ | small substructure |
| ☐ | algorithmic |
| ☐ | real |

Click to Save Answer & Move





CS502:Grand Quiz

Question # 25 of 30 (Start time: 01:47:48 PM, 27 June 2021)

Quick sort is based on _____ strategy.

Select the correct option

☐	Greedy approach
☐	Dynamic programming
☐	Graph Theory
☒	Divide-and-Conquer

Click t





Question # 26 of 30 (Start time: 01:48:13 PM, 27 June 2021)

_____ is one of the few problems, where provable lower bounds exist on how fast we can sort.

Select the correct option

☐	Graphing
☐	Searching
☒	Sorting
☐	Both Searching & Sorting

Click to Save Answer





CS502:Grand Quiz

Question # 27 of 30 (Start time: 01:48:42 PM, 27 June 2021)

In average-case time the probability of seeing input is denoted by_____.

Select the correct option

☐	$p< >$
☐	$p\{l\}$
☒	$p(l)$
☐	$p[l]$





CS502:Grand Quiz

Question # 28 of 30 (Start time: 01:49:09 PM, 27 June 2021)

What is the average running time of a quick sort algorithm?

Select the correct option

☐	$O(\log n)$
☒	$O(n \log n)$
☐	$O(n^2)$
☐	$O(n)$

Click to





CS502:Grand Quiz

Question # 29 of 30 (Start time: 01:49:41 PM, 27 June 2021)

Two functions $4n^2$ and $(8n^2 + 2n - 3)$ are _____.

Select the correct option

☐	hard to calculate
☐	related to power functions
☐	not equivalent
☒	asymptotically equivalent

Click to Save





CS502:Grand Quiz

Question # 30 of 30 (Start time: 01:50:06 PM, 27 June 2021)

Boolean operation is a _____ operation on an idealized RAM model of computation.

Select the correct option

☐	Starting
☒	Basic
☐	Advance
☐	Normal





CS502:Grand Quiz

Question # 1 of 30 (Start time: 02:09:38 PM, 27 June 2021)

In generating Fibonacci Sequence, we can avoid unnecessary repetitions by _____ process.

Select the correct option

☐	Tokenization
☐	Randomization
☐	Memorization
☒	Memoization

Click to Save





Question # 2 of 30 (Start time: 02:10:14 PM, 27 June 2021)

The O-notation is used to state only the asymptotic _____ bounds.

Select the correct option

☒	Upper
☐	Two
☐	Lower
☐	Both lower & upper

Click to Save





CS502:Grand Quiz

Question # 3 of 30 (Start time: 02:10:37 PM, 27 June 2021)

The important factors to measure the running time of the brute-force 2-d maxima algorithm are

Select the correct option

- | | |
|----------------------------------|--|
| ☒ | All of above |
| ☐ | The number of comparisons that are performed |
| ☐ | Count the number of times an element of P is accessed |
| ☐ | Count the number of steps of the pseudo code that are executed |



Question # 4 of 30 (Start time: 02:11:04 PM, 27 June 2021)

In asymptotical analysis of $n(n - 3)$ and $4n*n$, as n becomes large, the dominant (fastest growing) term is some constant times _____.

Select the correct option

☐	n
☒	$n*n$
☐	$n+1$
☐	$n-1$

Submit





CS502:Grand Quiz

Quiz Start Time: 02:09 PM

Question # 5 of 30 (Start time: 02:11:42 PM, 27 June 2021)

Consider two matrices A and B, where A is 10×30 and B is 30×40 . What is correct number of multiplications require to multiply A and B?

Select the correct option

[Reload M](#)

- | | |
|----------------------------------|--------------------------|
| ☐ | 10×30 |
| ☐ | 30×30 |
| ☒ | $10 \times 30 \times 40$ |
| ☐ | 30×40 |

[Click to Save Answer & Move to N](#)

CS502:Grand Quiz

Quiz Start Time: 02:09 PM,

Question # 6 of 30 (Start time: 02:12:16 PM, 27 June 2021)

In _____ involves breaking up the problem into sub problems whose solutions can be combined to solve the global problem.

Select the correct option

- | | |
|----------------------------------|------------------------------|
| ☒ | Dynamic programming solution |
| ☐ | Complexity Theory |
| ☐ | Greedy Algorithms |
| ☐ | Divide and Conquer Strategy |

[Click to Save Answer & Move to Nex](#)



Question # 7 of 30 (Start time: 02:12:37 PM, 27 June 2021)

If the indices passed to merge sort algorithm are _____, then this means that there is only one element to sort.

Select the correct option

☐	Small
☒	Equal
☐	Large
☐	Not Equal

Click to Save An





CS502:Grand Quiz

Question # 8 of 30 (Start time: 02:13:27 PM, 27 June 2021)

Recurrence can be described in terms of_____.

Select the correct option

- | | |
|----------------------------------|--------|
| ☐ | Graph |
| ☐ | Array |
| ☒ | Tree |
| ☐ | Linear |

Click to Save





Question # 9 of 30 (Start time: 02:13:55 PM, 27 June 2021)

What is the worst-case time of a quick sort which happens rarely?

Select the correct option

☐	$O(\log n)$
☐	$O(n \log n)$
☐	$O(n)$
☒	$O(n^2)$

Click to Save Answer





CS502:Grand Quiz

Question # 10 of 30 (Start time: 02:14:23 PM, 27 June 2021)

$f(n)$ and $g(n)$ are asymptotically equivalent. This means that they have essentially the same _____ for large n .

Select the correct option

☐	Variables
☒	Growth rates
☐	Results
☐	Size





CS502:Grand Quiz

Question # 11 of 30 (Start time: 02:14:54 PM, 27 June 2021)

We can multiply two matrices A and B only when they are compatible which means:

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | Number of rows in A must be equal to number of rows in B |
| ☐ | Number of columns in A must be equal to number of columns in B |
| ☒ | Number of columns in A must be equal to number of rows in B |
| ☐ | Number of rows and columns do not matter |

Click to





CS502:Grand Quiz

Question # 12 of 30 (Start time: 02:15:21 PM, 27 June 2021)

A Principal operation for maintaining the heap property is called heapify, it is also called:

Select the correct option

☐	Sifting Left
☒	Sifting Down
☐	Sifting Up
☐	Sifting Right

Click to Save





CS502:Grand Quiz

Question # 13 of 30 (Start time: 02:15:48 PM, 27 June 2021)

While applying the Sieve technique to selection sort, how to choose a pivot element.

Select the correct option

☐	Sequentially
☐	Linear
☐	Through mean
☒	Randomly

Click to Save





Question # 14 of 30 (Start time: 02:16:17 PM, 27 June 2021)

In asymptotical analysis of $n^*(5 + 2) - 3$, as n becomes large, the dominant (fastest growing) term is some constant times _____

Select the correct option

☒	n
☐	n+1
☐	n_1
☐	n*n





CS502:Grand Quiz

Question # 15 of 30 (Start time: 02:16:39 PM, 27 June 2021)

Which one of the following sorting algorithms is the fastest?

Select the correct option

☐	Heap sort
☒	Quick sort
☐	Merge sort
☐	Insertion sort

Click to





CS502:Grand Quiz

Question # 16 of 30 (Start time: 02:17:03 PM, 27 June 2021)

Which type of operations does RAM perform?

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☐ | Access element of array |
| ☐ | Comparison or Boolean |
| ☒ | All of above |
| ☐ | Assign value to a variable |

Click





CS502:Grand Quiz

Question # 17 of 30 (Start time: 02:17:30 PM, 27 June 2021)

While analyzing algorithms, _____ and _____ are usually considered difficult to calculate.

Select the correct option

☐	Finite, Infinite
☒	Floor, ceiling
☐	Row, Column
☐	Graph, Tree



CS502:Grand Quiz

Question # 18 of 30 (Start time: 02:17:59 PM, 27 June 2021)

In Selection problem, the Sieve technique works in _____

Select the correct option

☒	Phases
☐	Constant time
☐	Non-recursive manner
☐	One complete go





Question # 19 of 30 (Start time: 02:18:32 PM, 27 June 2021)

In the analysis of Selection algorithm, we get the convergent _____ series.

Select the correct option

☐	Linear
☒	Geometric
☐	Arithmetic
☐	Harmonic

Click to Save Answer





CS502:Grand Quiz

Question # 20 of 30 (Start time: 02:18:55 PM, 27 June 2021)

In the analysis of algorithms,_____ plays an important role.

Select the correct option

☐	money
☐	time
☒	growth rate
☐	text analysis





CS502:Grand Quiz

Question # 21 of 30 (Start time: 02:19:20 PM, 27 June 2021)

Boolean operation is a _____ operation on an idealized RAM model of computation.

Select the correct option

☐	Normal
☐	Starting
☒	Basic
☐	Advance





CS502:Grand Quiz

Question # 22 of 30 (Start time: 02:20:01 PM, 27 June 2021)

In the statement
"if $(P[i].x < P[j].x) \ \& \ (P[i].y < P[j].y)$ ",
the number of times elements of P are accessed is _____

Select the correct option

☐	3
☐	1
☒	4
☐	2



CS502:Grand Quiz

Question # 23 of 30 (Start time: 02:20:40 PM, 27 June 2021)

After partitioning array in Quick sort, pivot is placed in a position such that

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | Pivot is the first element of array |
| ☐ | Pivot is the last element of array |
| ☐ | Values larger than pivot are on left and smaller than pivot are on right |
| ☒ | Values smaller than pivot are on left and larger than pivot are on right |

Click to





CS502:Grand Quiz

Question # 24 of 30 (Start time: 02:21:07 PM, 27 June 2021)

If there are duplicates in _____ sort, then each bin can be replaced by a linked list.

Select the correct option

☐	Merge
☐	Quick
☐	Heap
☒	Bucket

Click to





CS502:Grand Quiz

Question # 25 of 30 (Start time: 02:21:34 PM, 27 June 2021)

If the time complexity of an algorithm is given by $O(1)$, then its time complexity would be:

Select the correct option

☐	Exponential
☐	Polynomial
☐	Average
☒	Constant





CS502:Grand Quiz

Question # 26 of 30 (Start time: 02:22:09 PM, 27 June 2021)

In Selection problem, the Sieve technique _____

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | Do not uses Divide and Conquer approach |
| ☐ | Do not work recursively |
| ☐ | Adds some more input items each time |
| ☒ | Eliminates undesired data items each time |

Click to





CS502:Grand Quiz

Question # 27 of 30 (Start time: 02:22:34 PM, 27 June 2021)

In Dynamic Programming, our approach is to _____.

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | Express the problem non-recursively |
| ☐ | Input several sub-problems simultaneously |
| ☐ | Develop the solution in a top-down fashion |
| ☒ | Build the solution in a bottom-up fashion |



CS502:Grand Quiz

Question # 28 of 30 (**Start time: 02:23:04 PM, 27 June 2021**)

_____ algorithm based on Dynamic Programming strategy.

Select the correct option

☐	Quick Sort
☐	Binary Tree
☒	Edit distance
☐	Heap Sort





Question # 29 of 30 (Start time: 02:23:31 PM, 27 June 2021)

The worst case running time of quick sort algorithm _____.

Select the correct option

☒	Is quadratic
☐	Is linear
☐	Is always Exponential
☐	Cannot be quadratic

Click to Save Answer





CS502:Grand Quiz

Question # 30 of 30 (Start time: 02:23:59 PM, 27 June 2021)

The process of _____ ends when you are left with such tiny pieces remaining that it is trivial to solve them.

Select the correct option

☐	Axis-sweep
☐	Plan-sweep
☒	Divide and Conquer
☐	Brute-force

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